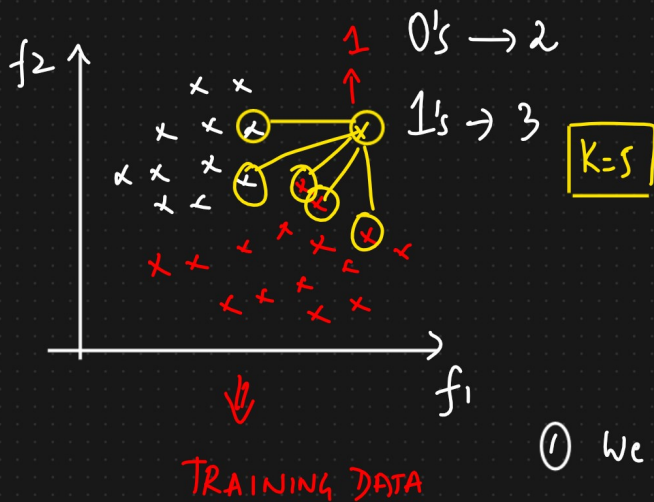


K Nearest Neighbour (KNN)

① Classification

② Regression

① Classification



f_1	f_2	y	{Binary Categories}
-	-	0	{Multi class}
		1	
		1	
		0	

① We have to initialize the K value

$$K \geq 0$$

$K=1, 2, 3, 4, 5, 6 \Rightarrow$ Hyperparameter

$K=5$

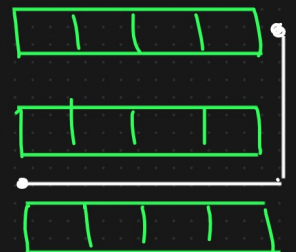
② Find the K Nearest Neighbour from the Test DATA

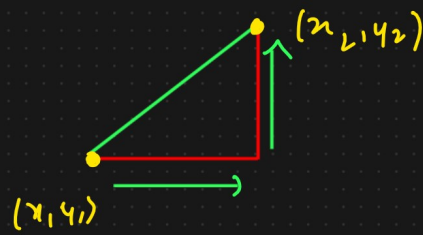
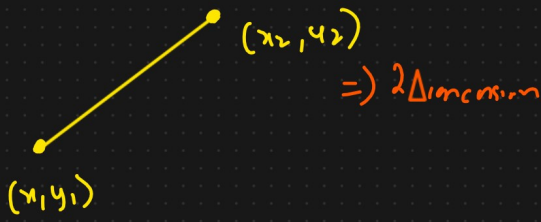
③ From those $K=5$ how many neighbours belongs to 0 category and 1 category.

Distance Metrics

① Eucledian Distance

② Manhattan Distance

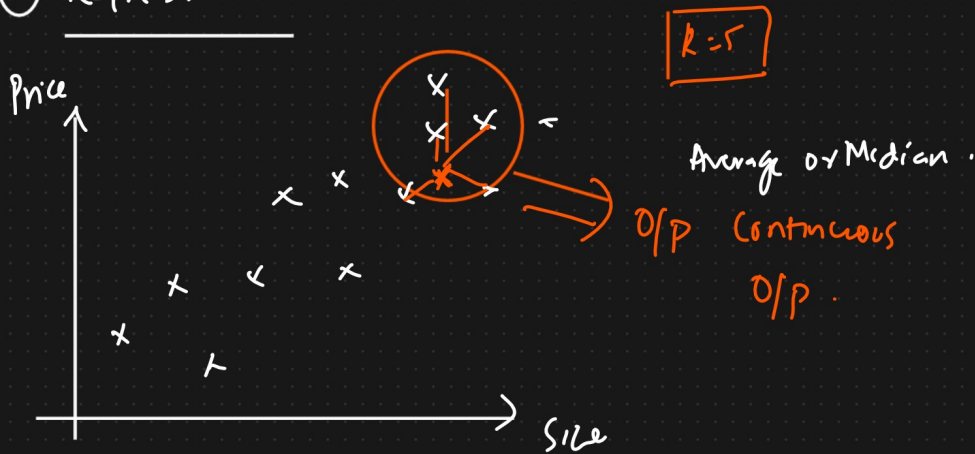




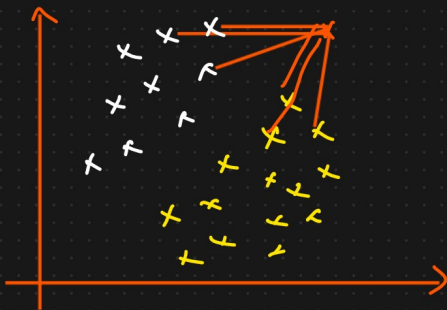
$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

② Regression



Variant of KNN



Time Complexity $\uparrow\uparrow\uparrow$

$O(n)$

- ① KD Tree
 - ② Ball Tree
- } Binary Tree



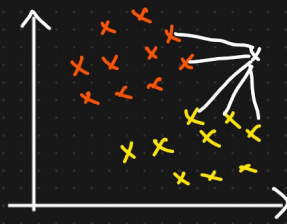
Time complexity $\downarrow\downarrow$.

Variants of KNN

Two Variants

- ① KD Tree
- ② Ball Tree

Binary Tree



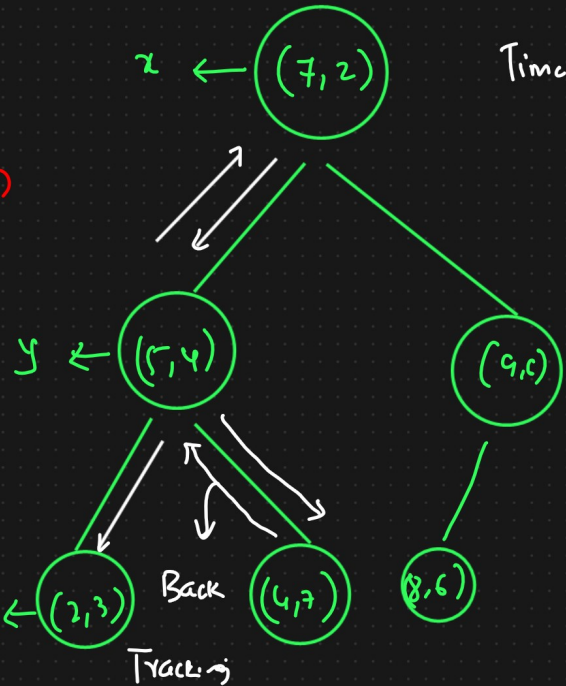
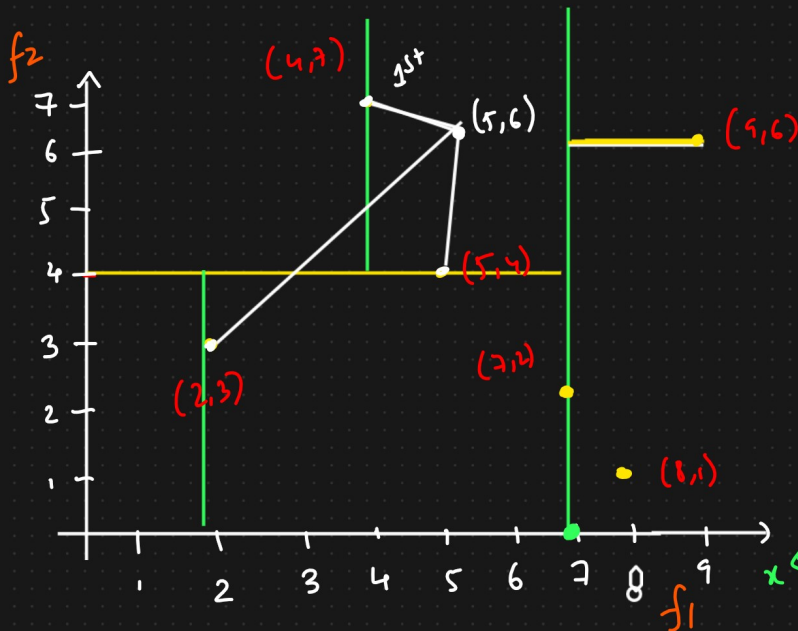
$K=3$

Time Complexity

Brute Search

Auto

① KD Tree $\Rightarrow$ K-Dimensions Tree



Time Complexity
 $\downarrow \downarrow$

① Median of the x coordinates

2, 4, 5, 7, 8, 9

$$\downarrow$$

$$\frac{5+7}{2} = 6.5 \Rightarrow 7$$

② Median of y axis

1, 2, 3.4, 6, 7

② Ball Tree

